

FULL STACK DEVELOPER – INTERVIEW ASSIGNMENT

Project: Revamp Riyadhvi Software Technologies Website

Project Type

Dynamic Multi-Page Website + 3D Interactive Experience + Full-Stack Development + AI-Assisted Development

1. PROJECT OVERVIEW

Riyadvi Software Technologies is looking to revamp its existing corporate website into a **premium, modern, dynamic and interactive digital experience**.

The objective of this assignment is to evaluate the candidate's ability to combine:

- Full-stack web development
- Modern UI/UX
- 3D web technologies
- Advanced animations
- AI-assisted development
- Dynamic content architecture
- Backend/API development
- Database integration
- Responsive development
- Performance optimization

Existing Website

Use the existing Riyadhvi website as a reference for company information and content:

www.riyadvisoftwaretechnologies.com

The candidate may improve, restructure and rewrite the presentation of the content where appropriate.

2. IMPORTANT – THIS IS NOT A LANDING PAGE

This assignment is **NOT** a single-page landing-page project.

The candidate must build a **complete dynamic corporate website** with:

- Multiple pages
- Proper routing

- Reusable components
- Dynamic content
- Individual service pages
- Individual portfolio case studies
- Individual blog pages
- Individual career/job pages
- Backend APIs
- Database integration
- Functional forms
- Lead management
- Responsive design

The homepage is only one part of the website.

The final result should resemble a **real production website architecture** that can continue to grow after development.

3. PRIMARY BUSINESS OBJECTIVE

The website should position Riyadvi as:

A Technology & Digital Solutions Partner – not simply a software development vendor.

The website should communicate:

- Technology expertise
- Innovation
- Business understanding
- Digital transformation
- Design excellence
- Long-term partnership
- Business growth

The website should encourage visitors to:

- Book a consultation
 - Contact Riyadvi
 - Request a quote
 - Download a lead magnet
 - Submit a Business Health Checkup
 - Explore services
 - View portfolio/case studies
-

4. DESIGN DIRECTION

The overall experience should be:

Premium + Futuristic + Interactive + 3D + Professional

Avoid a generic corporate website.

The candidate should create a distinctive visual language using:

- 3D objects
- Depth
- Parallax
- Interactive elements
- Scroll-based storytelling
- Cinematic transitions
- Smooth scrolling
- Advanced typography
- Micro-interactions
- Glass / metallic surfaces where appropriate
- Particles / digital effects where appropriate

The design should feel like a **modern technology company**, while remaining professional and usable.

Inspiration

Candidates may research modern digital experiences and resources such as:

- Turing
- InVision
- CRED
- getlayers.ai
- threeui.com
- Aceternity UI
- Magic UI
- Shadcn/UI
- Three.js examples
- React Three Fiber ecosystem

These are references for inspiration, not templates to copy.

5. BRANDING

Primary Colors

Gold: #D4AF37

Black: #000000

The candidate may introduce suitable neutral shades, gradients and supporting colors where required for accessibility and visual hierarchy.

Fonts

Use one or more of:

- Inter
- Poppins
- Montserrat

Tone

- Premium
 - Professional
 - Innovative
 - Trustworthy
 - Technology-focused
 - Business-oriented
-

6. AI-ASSISTED DEVELOPMENT – MANDATORY

One of the main objectives of this assignment is to evaluate the candidate's ability to use **AI effectively as part of a modern development workflow**.

Candidates are encouraged to use AI tools such as:

- ChatGPT
- Claude
- Gemini
- Cursor
- GitHub Copilot
- Lovable
- v0
- Bolt
- Replit
- Midjourney
- Leonardo AI
- Adobe Firefly
- Other relevant AI development/design tools

The candidate should use AI to accelerate development, research, design exploration, coding, debugging, asset generation or documentation.

However:

The candidate must understand and customize the generated output.

We are NOT evaluating:

"Who can generate a website using AI?"

We are evaluating:

"Who can use AI effectively while still understanding, controlling, debugging and improving the final product?"

7. AI USAGE DOCUMENTATION

The README must contain an:

AI Tools Used

section.

For each major AI tool used, explain:

- Tool name
- Purpose
- Example prompt
- What was generated
- What was manually changed
- Why the tool was selected

Example:

AI Tool:
Claude

Purpose:
Generate an initial React Three Fiber concept for the hero section.

Prompt:
Create an interactive 3D digital ecosystem representing technology and business transformation.

Manual Work:
The generated implementation was modified, optimized for performance and integrated into the Riyadhvi design system.

8. 3D / ADVANCED ANIMATION – MANDATORY

The website must contain **meaningful 3D and advanced animation experiences**.

At least **4 technologies** from the following list should be meaningfully implemented:

Technology	Possible Use
Three.js	Real-time 3D
React Three Fiber	Three.js inside React
Drei	React Three Fiber helpers
GSAP	Advanced animation/timelines
GSAP ScrollTrigger	Scroll-based storytelling
Lenis	Smooth scrolling
Rive	Interactive animations
Lottie / dotLottie	Lightweight animated illustrations
Motion	UI/micro animations
Anime.js	Lightweight animation
Theatre.js	Complex animation timelines
Blender	Custom 3D models/animations
Spline	Interactive 3D experiences

The candidate does **not** need to use every technology.

Important

Installing a library is not considered implementation.

The selected technologies must visibly contribute to the website experience.

9. 3D SHOULD NOT BE LIMITED TO THE HERO

The 3D experience should be incorporated throughout the website where appropriate.

At least **three major sections/pages** should contain meaningful interactive visual experiences.

Examples:

Homepage

Interactive 3D technology ecosystem

Services

3D visualization for individual services

Portfolio

Interactive 3D case-study presentation

About

Interactive company timeline

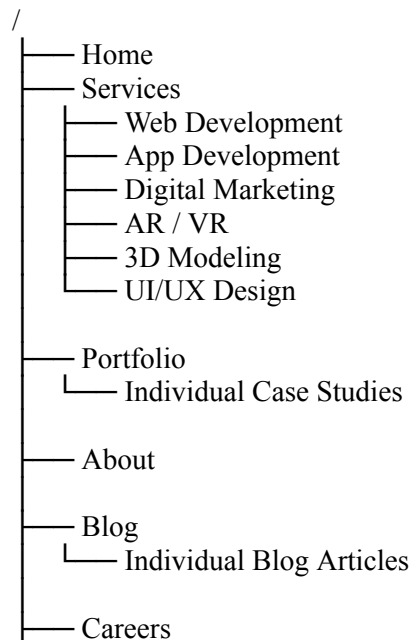
Technology

3D technology ecosystem

The candidate can propose their own creative implementation.

10. WEBSITE STRUCTURE

The minimum required structure is:



└─ Individual Job Details
└─ Contact
└─ Business Health Checkup
└─ Software Project Planning Guide

11. HOMEPAGE

The homepage should be the primary showcase of the candidate's design and technical ability.

Hero Section

Headline:

Custom Software & Digital Solutions to Grow Your Business

Supporting text:

Web & App Development, UI/UX Design, and Business Strategy – all tailored to your needs.

Primary CTA:

Book a Free Consultation

Secondary CTA:

Explore Our Solutions

3D Hero

Create an interactive 3D experience representing:

- Technology
- Digital transformation
- Connected systems
- Business growth
- Innovation

Possible concepts:

- Digital ecosystem
- Connected 3D nodes
- Interactive digital sphere
- 3D data network
- Abstract technology structure

- Metallic digital object
- Software architecture visualization

The candidate can create the asset using:

- Three.js
- React Three Fiber
- Blender
- Spline
- AI-generated assets
- Other appropriate technologies

The hero should respond appropriately to interaction such as:

- Cursor movement
 - Hover
 - Scroll
 - Device movement where appropriate
-

12. DIGITAL TRANSFORMATION SECTION

Create an interactive section explaining Riyadvi's approach.

For example:

Business Challenge



Strategy



Design



Technology



Launch



Growth

Use scroll-based animation or interactive storytelling.

Recommended technologies:

- GSAP
- ScrollTrigger
- Lenis
- Motion
- Three.js

13. SERVICES

Display the six core services:

1. Web Development
2. App Development
3. Digital Marketing
4. AR / VR
5. 3D Modeling
6. UI/UX Design

Do not make these ordinary static cards.

Create an interactive visual treatment for each service.

Examples:

Web Development

Interactive browser / web environment

App Development

3D smartphone/device

Digital Marketing

Animated data/growth visualization

AR / VR

3D immersive environment

3D Modeling

Interactive rotatable 3D object

UI/UX

Floating interface elements

Each service should link to its own service page.

14. TECHNOLOGY ECOSYSTEM

Create an interactive section showcasing technologies.

Examples:

- React
- Next.js
- Node.js
- MongoDB
- MySQL
- JavaScript
- Three.js
- React Three Fiber
- WordPress
- Other relevant technologies

Possible implementations:

- 3D technology constellation
 - Orbiting ecosystem
 - Interactive sphere
 - Animated technology network
 - Floating logos
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15. WHY RIYADVI

Include:

Since 2021

Show the company journey using an interactive timeline.

Business Health Checkup

Explain Riyadvi's approach to identifying digital/business improvement opportunities.

End-to-End Solutions

Show the complete journey:

Strategy



Design



Development



Marketing

↓
Optimization
↓
Growth

16. BUSINESS HEALTH CHECKUP

Create a dedicated lead-generation experience.

Page:

[/business-health-checkup](#)

Headline:

Is Your Business Ready for Its Next Digital Growth Stage?

Include an interactive multi-step form.

Possible steps:

1. Business Information
2. Website & Digital Presence
3. Marketing
4. Technology
5. Business Challenges
6. Submission

The submitted information must be sent to the backend and stored in the database.

17. LEAD MAGNET

Create a dedicated page:

[/software-project-planning-guide](#)

Title:

Download Software Project Planning Guide

Collect:

- Name
- Company
- Email

- Phone

On submission:

- Validate the form
 - Send data to backend
 - Store lead
 - Provide download/access confirmation
-

18. PORTFOLIO

Page:

[/portfolio](#)

Create a premium dynamic portfolio experience.

Possible projects:

- Puratap
- Wanaromah Perfumers
- Laxmi Astro AI
- Tony & Guy
- Studio11
- Sivam Physio Care
- Pearl Housing
- Nugenica Biotech Lab
- VisDoc
- Cube Dental

Portfolio cards should lead to dynamic case studies.

Example:

[/portfolio/puratap](#)
[/portfolio/wanaromah](#)
[/portfolio/laxmi-astro-ai](#)

Each case study should contain:

- Client
- Industry
- Challenge
- Solution
- Technologies
- Results
- Visuals

- Related services

At least one case study should use an interactive/3D presentation.

19. SERVICE PAGES

Create individual dynamic service pages.

Required:

/services/web-development
/services/app-development
/services/digital-marketing
/services/ar-vr
/services/3d-modeling
/services/ui-ux-design

Each page should contain:

- Interactive hero
- Problem
- Solution
- Key features
- Industry use cases
- Technology stack
- Process
- Related portfolio
- CTA

CTA:

Get a Quote

Use a reusable service-page architecture rather than creating six unrelated pages.

20. ABOUT PAGE

Page:

/about

Include:

- Company story

- Founded in 2021
- Vision
- Mission
- Values
- Approach
- Awards/recognition
- Company milestones
- Optional team

Create an engaging visual presentation using animation, timeline or 3D elements.

21. BLOG

Page:

`/blog`

Individual article:

`/blog/[slug]`

The blog must be structured dynamically.

Include:

- Featured article
- Categories
- Tags
- Search
- Article cards
- Individual article pages
- Related articles

The architecture should allow future integration with:

- Strapi
 - Sanity
 - WordPress
 - Database
 - Other CMS/API
-

22. CAREERS

Page:

[/careers](#)

Individual job:

[/careers/\[job-slug\]](#)

Include:

- Job listings
- Department filters
- Designation filters
- Experience filters
- Job details
- Responsibilities
- Requirements
- Apply button

Application form:

- Name
- Email
- Phone
- Position
- Resume
- Message

Application data must be handled by the backend.

23. CONTACT

Page:

[/contact](#)

Include:

- Contact form
- Name
- Email
- Phone
- Company
- Requirement
- Message

Integrations:

- Backend API
- Database

- Email notification
- WhatsApp
- Calendly

Optional:

- Location/map
-

24. DYNAMIC CONTENT ARCHITECTURE

The website should be built using reusable structures.

For example:

Service Data



Reusable Service Template



Web Development

App Development

Digital Marketing

AR/VR

3D Modeling

UI/UX

Similarly:

Portfolio Data



Reusable Case Study Template



Project A

Project B

Project C

And:

Job Data



Reusable Job Template



Job A

Job B

Job C

Avoid creating completely separate hardcoded components for every item.

25. BACKEND

The project must include a functional backend.

Preferred:

Node.js + Express.js

Other acceptable options:

- Laravel
 - Django
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26. DATABASE

Use one:

- MongoDB
- PostgreSQL
- MySQL

At minimum, store:

Contact Enquiries

Consultation Requests

Business Health Checkup Leads

Lead Magnet Leads

Career Applications

Where practical, structure:

- Services
- Portfolio
- Blog
- Careers

as dynamic data models or API-ready structures.

27. API REQUIREMENTS

Example endpoints:

POST /api/contact

POST /api/consultation

POST /api/health-checkup

POST /api/lead-magnet

POST /api/applications

The API should include:

- Validation
 - Error handling
 - Proper response structure
 - Environment variables for sensitive configuration
-

28. ADMIN DASHBOARD

A basic admin dashboard is strongly preferred.

The dashboard should allow viewing:

Dashboard

- Total enquiries
- Consultation requests
- Health Checkup leads
- Lead Magnet leads
- Job applications

Enquiries

Show:

- Name
- Email
- Phone
- Company
- Requirement

- Date
- Status

Careers

View submitted applications.

The dashboard does not need to be a full enterprise CRM.

The objective is to demonstrate full-stack understanding.

29. RESPONSIVE DESIGN

The website must work properly on:

- Desktop
- Laptop
- Tablet
- Mobile

The 3D experience must have a mobile strategy.

If a 3D scene is too heavy for mobile, implement:

- Reduced rendering
 - Reduced animation
 - Lightweight alternative
 - Static fallback
 - Device-specific optimization
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30. PERFORMANCE

The candidate must consider:

- Lazy loading
- Code splitting
- Image optimization
- 3D asset optimization
- Texture optimization
- Font optimization
- Animation performance
- Mobile GPU limitations
- Lighthouse performance

The website should not sacrifice usability for visual effects.

31. TECHNOLOGY STACK

Frontend

Preferred:

- React.js / Next.js
- TailwindCSS / Styled Components

Backend

- Node.js + Express
- Laravel
- Django

Database

- MongoDB
- PostgreSQL
- MySQL

Deployment

Frontend:

- Vercel preferred

Backend:

- Render
- Railway
- Other suitable platform

32. GIT REQUIREMENTS

Provide a GitHub repository containing:

/frontend
/backend
/README.md

Use meaningful commit history.

Examples:

feat: implement interactive 3D hero
feat: create dynamic service pages
feat: implement portfolio case study architecture
feat: add contact API
feat: integrate MongoDB
feat: add business health checkup
perf: optimize 3D assets
fix: improve mobile navigation

Avoid submitting the entire project using only:

Initial commit

33. README REQUIREMENTS

The README must contain:

- Project overview
 - Features
 - Tech stack
 - Installation instructions
 - Environment variables
 - Database setup
 - API endpoints
 - Deployment instructions
 - AI tools used
 - 3D libraries used
 - Animation libraries used
 - Third-party assets
 - Performance optimization
 - Known limitations
 - Future improvements
-

34. DEPLOYMENT

A live deployment is **mandatory**.

Provide:

Frontend

Vercel deployment URL

Backend

Live backend/API URL

The evaluator must be able to access the project without running it locally.

35. TIME LIMIT

Development Time: 3–5 Days

The candidate is not expected to build a huge enterprise website within this time.

The objective is to demonstrate **quality, technical ability and decision-making**.

Priority Order

If time becomes limited, prioritize:

1. Homepage
2. 3D Hero
3. Services + dynamic service architecture
4. Portfolio + dynamic case study architecture
5. Lead-generation functionality
6. Backend + database
7. Responsive experience
8. Additional pages

Blog and Careers may contain limited sample content, but the architecture should be scalable.

36. WHAT WE ARE NOT LOOKING FOR

Do NOT submit:

- A single landing page
 - One long scrolling page
 - Static service cards without individual pages
 - Static portfolio cards without case studies
 - Fake contact forms
 - Frontend-only implementation
 - Random 3D effects with no purpose
 - Excessive animations that make the site slow
 - Unmodified AI-generated templates
 - Copy-pasted websites
 - A project with no meaningful Git history
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37. WHAT WE ARE LOOKING FOR

We are looking for a developer who can combine:

- **AI**
- **3D**
- **Animation**
- **UI/UX**
- **Frontend**
- **Backend**
- **Database**
- **Business Understanding**

into one cohesive website.

The candidate should demonstrate the ability to:

- Research modern technologies
 - Select appropriate libraries
 - Use AI effectively
 - Understand AI-generated code
 - Build reusable components
 - Create dynamic pages
 - Build APIs
 - Connect databases
 - Optimize 3D experiences
 - Build responsive interfaces
 - Deploy a working application
-

38. FINAL INTERVIEW DEMONSTRATION

After submission, the candidate should be prepared to give a **10–15 minute technical walkthrough**.

The candidate should explain:

1. Design

Why was this design direction selected?

2. 3D

How was the 3D experience implemented?

3. AI

Which AI tools were used and why?

4. Development

Which parts were generated using AI and which were manually developed?

5. Architecture

How are the frontend, backend and database connected?

6. Dynamic Content

How would the company add another service, project, blog article or job without rebuilding the page?

7. Performance

How was the 3D experience optimized?

8. Challenges

What were the biggest technical challenges?

9. Production

What would you improve if you had another 1–2 weeks?

39. EVALUATION CRITERIA

Area	Weight
3D & Interactive Experience	30%
AI-Assisted Development	15%
UI/UX & Visual Quality	15%
Frontend Architecture	10%
Backend & Database	10%
Performance & Responsiveness	10%
Git, Documentation & Deployment	10%
Total	100%

3D & Interactive Experience – 30%

Evaluate:

- Quality of 3D
- Meaningful implementation
- Interactivity
- Animation
- Creativity
- Performance

AI-Assisted Development – 15%

Evaluate:

- AI tool selection
- Prompt quality
- Ability to modify generated code
- Debugging
- Understanding of AI-generated output
- Appropriate use of AI

UI/UX – 15%

Evaluate:

- Visual quality
- Navigation
- User experience
- Responsive design
- Conversion flow
- Consistency

Frontend – 10%

Evaluate:

- React/Next.js
- Component architecture
- Reusability
- Routing
- State management
- Code quality

Backend – 10%

Evaluate:

- API architecture
- Database integration
- Validation
- Error handling

- Form processing

Performance – 10%

Evaluate:

- Page loading
- 3D optimization
- Asset optimization
- Mobile performance
- Animation performance

Git / Documentation / Deployment – 10%

Evaluate:

- Commit history
 - README
 - Environment setup
 - Deployment
 - AI documentation
-

40. FINAL EXPECTATION

The final website should not feel like an interview assignment.

It should feel like the beginning of a **real premium Riyadvi corporate website**.

The final experience should communicate:

Riyadvi is a technology and digital solutions partner that combines technology, design, innovation and business thinking to help businesses grow.

The core challenge is to demonstrate:

AI + 3D + Animation + Full-Stack Development + Dynamic Architecture + Premium UI/UX

within a practical 3–5 days development window.